

Semester 1 Exam

Baldivis
Secondary College

Resources allowed: Calculator and a handwritten A4 notes page double sided

Name: T

Class: _____

Time: 50 minutes

Total Marks: 50

1) [5 Marks]

Simplify the following leaving your answers with positive indices.

a) $7a^2 \times 4a$ $28a^3$

b) $\frac{m^9}{m^3}$ m^6

c) $(h^5)^2$ h^{10}

d) $4x^0$ 4

e) $10p^{-5}$ $\frac{10}{p^5}$

2. [4 marks]

Simplify each of the expressions:

a) $12j + 3k - 9j + k$
 $3j + 4k$

b) $7yz \times 3z$ $21yz^2$

c) $20q - 14pq + 5p - qp$
 $20q + 5p - 15pq$

d) $\frac{20bc}{25a}$ $\frac{4b}{5}$

3. [4 Marks]

Expand and simplify each of the expressions:

a) Expand and simplify

(i) $5(9g - 4)$
 $45g - 20$

ii) $8 + 7(10 - 3w)$

$8 + 70 - 21w$
 $78 - 21w$

b) Factorise

(i) $7x + 21$
 $7(x + 3)$

(ii) $20a^2b - 15a$

$5a(4ab - 3)$

4. [2 marks]

Multiple choice: Circle your answer clearly

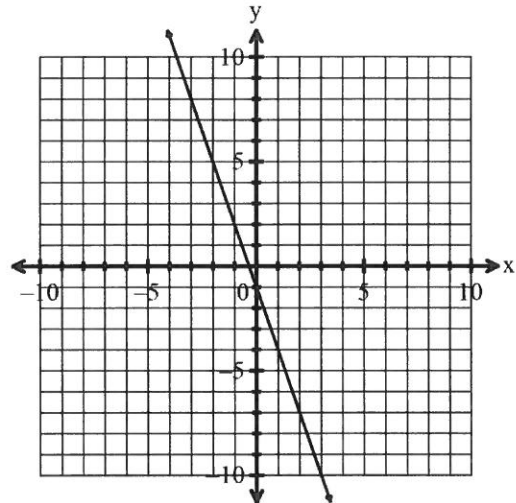
i. The slope of the given line is:

a) 3

☒ b) -3

c) $\frac{1}{3}$

d) $-\frac{1}{3}$



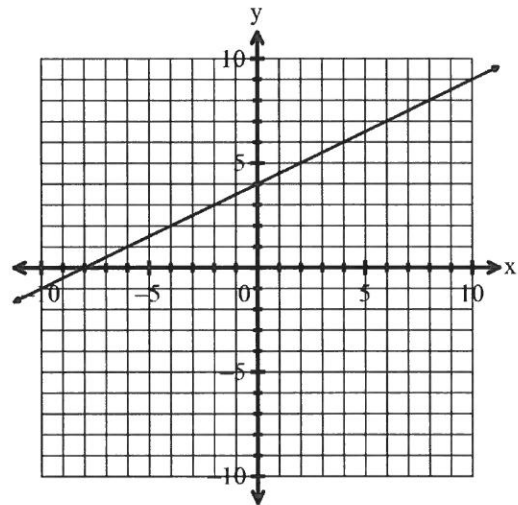
ii. The equation of the line is:

a) $y = -\frac{1}{2}x + 4$

b) $y = \frac{x}{2} - 4$

☒ c) $y = \frac{1}{2}x + 4$

d) $y = 2x - 4$



5. [3 Marks]

a) Plot the two points

$A(3, -3)$ and $B(5, -9)$ ✓

b) Use Pythagoras to find the distance between the two points

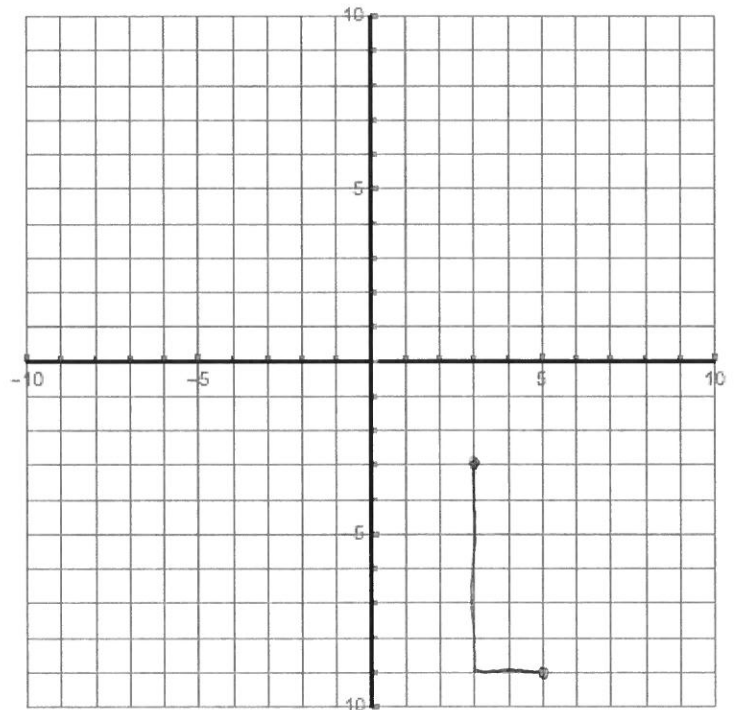
$$c^2 = a^2 + b^2$$

$$c^2 = 2^2 + 6^2$$

$$c^2 = 4 + 36$$

$$c = \sqrt{40}$$

$$c = 6.32$$

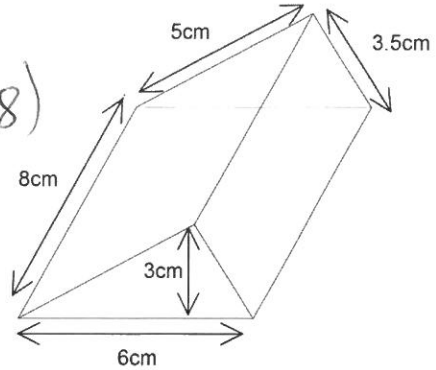


6. [4 marks]

Find the **surface area** of the following solids:

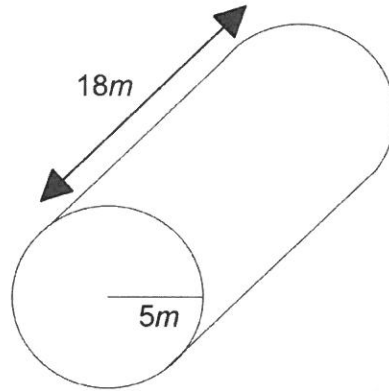
(a)

$$\begin{aligned} & \frac{1}{2} \times 6 \times 8 \times 2 + 6 \times 5 + 6 \times 8 + 6 \times 8 \\ &= \left(\frac{1}{2} \times 6 \times 8 \times 2 \right) + (5 \times 8) + (8 \times 3.5) + (6 \times 8) \\ &= 18 + 40 + 28 + 48 \\ &= 134 \text{ cm}^2 \end{aligned}$$



(b)

$$\begin{aligned} & 2 \times \pi \times r^2 + 2 \times \pi \times r \times l \\ &= 2 \times \pi \times 5^2 + 2 \times \pi \times 5 \times 18 \\ &= 157.5 + 565.5 \\ &= \underline{723.57 \text{ m}^2} \end{aligned}$$



7. [3 Marks]

Write these ratios in their simplest form

a) 25 : 50

$$1 : 2$$

b) 50 c : \$1

$$1 : 2$$

c) 12.5 : 10 : 5

$$\begin{aligned} & 25 : 20 : 10 \\ &= 5 : 4 : 2 \end{aligned}$$

8. [2 Marks]

350g of flour is to be divided into a ratio of 3:7 to be used in two different recipes. Find the smaller amount of flour.

$$(350 \div 10) \times 3 = \underline{105 \text{ g}}$$

9. [5 Marks]

Convert the following into rates with the units indicated

\$250 in 5 hours	\$ <u>50</u> /hr
12 truckloads in 3 hours	<u>4</u> truckloads per hour
468 people per year	<u>9</u> people per week
2.5 kg of beef cost \$15	\$ <u>6</u> /kg
18 L in 2 hours 18 000 mL in 7200s	<u>2.5</u> mL/s

10. [3 Marks]

- a) 1 Australian dollar (1AUD) can be exchanged for 1.05 New Zealand dollars (1.05NZD). How many New Zealand Dollars could you get for 60 Australian Dollars?

$$60 \times 1.05 = \$63 \quad \checkmark$$

- b) Tenaya is going on holiday and wants to buy a souvenir for her three best friends. Each one costs 6 New Zealand Dollars, and she only has 18 Australian dollars left. Can she afford these, and if so how much money in Australian dollars would she have left over?

$$6 \div 1.05 = \$5.71 \text{ NZ}$$

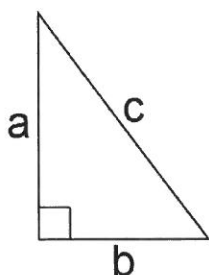
$$18 - (5.71 \times 3) = \$0.86 \text{ leftover} \quad //$$

11. [5 Marks]

(a) Complete the statements:

- (i) Pythagoras' theorem is only true for right-angled triangles.
- (ii) The longest side of a right angled triangle is called the hypotenuse.
- (iii) The sum of the squares of the shorter sides of a right angled triangle is equal to the square of the longer side / hypotenuse

(b) Write an equation to show how the three sides of Triangle 1 relate to each other.

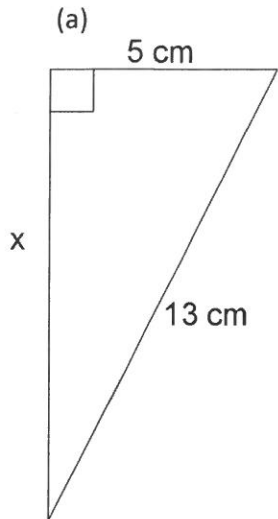


Triangle 1

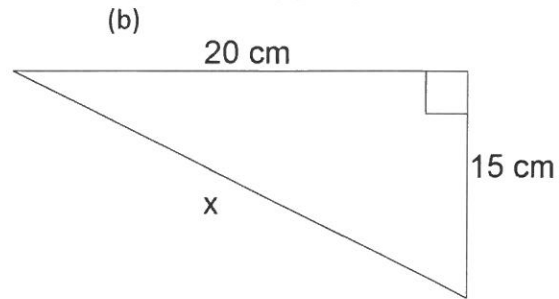
$$c^2 = a^2 + b^2$$

10. [7 marks]

Find the value of the variable in each of the following diagrams. Show your working giving accurate answers.

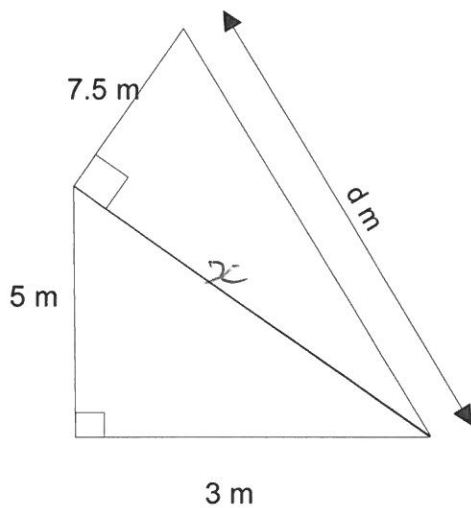


$$\begin{aligned}
 b^2 &= c^2 - a^2 \\
 x^2 &= 13^2 - 5^2 \\
 x^2 &= 169 - 25 \\
 x &= \sqrt{144} \\
 x &= 12 \text{ cm}
 \end{aligned}$$



$$\begin{aligned}
 c^2 &= b^2 + a^2 \\
 x^2 &= 20^2 + 15^2 \\
 x^2 &= 400 + 225 \\
 x &= \sqrt{625} \\
 x &= 25 \text{ cm}
 \end{aligned}$$

c)



$$\begin{aligned}
 c^2 &= a^2 + b^2 \\
 x^2 &= 5^2 + 3^2 \\
 x^2 &= 25 + 9 \\
 x &= \sqrt{34} \\
 x &= 5.83 \text{ m}
 \end{aligned}$$

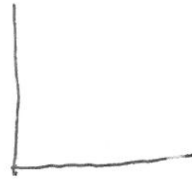
$$\begin{aligned}
 c^2 &= a^2 + b^2 \\
 d^2 &= 7.5^2 + 5.83^2 \\
 d^2 &= 56 + 34 \\
 d &= \sqrt{90} \\
 d &= 9.49 \text{ m}
 \end{aligned}$$

11. [4 Marks]

John leaves his farm and is driving West at a speed of 80 km/hr for 1.5 hours. He then turns right and travels North for 45 minutes at a speed of 110 km/hr.

- a) How far did he travel to the North?

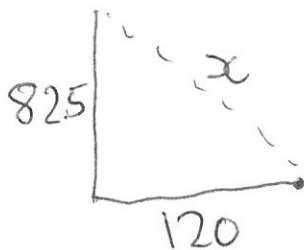
$$110 \times 0.75 \\ = 82.5 \text{ km}$$



- b) How far did he travel to the West?

$$80 \times 1.5 \\ = 120 \text{ km}$$

- c) His emergency radio has a range of 150km. Can he still send a message home? Show by how far he can or can't.



$$c^2 = a^2 + b^2 \\ x^2 = 82.5^2 + 120^2 \\ x^2 = 21\,206.25 \\ x = \sqrt{21\,206} \\ x = 145.62 \text{ km}$$

He can by 4.38 km